



Molecular weight distribution

Linear polyesters of
itaconic acid, propylene
glycol and succinic acid

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Collaboration with Cambridge University
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Overview

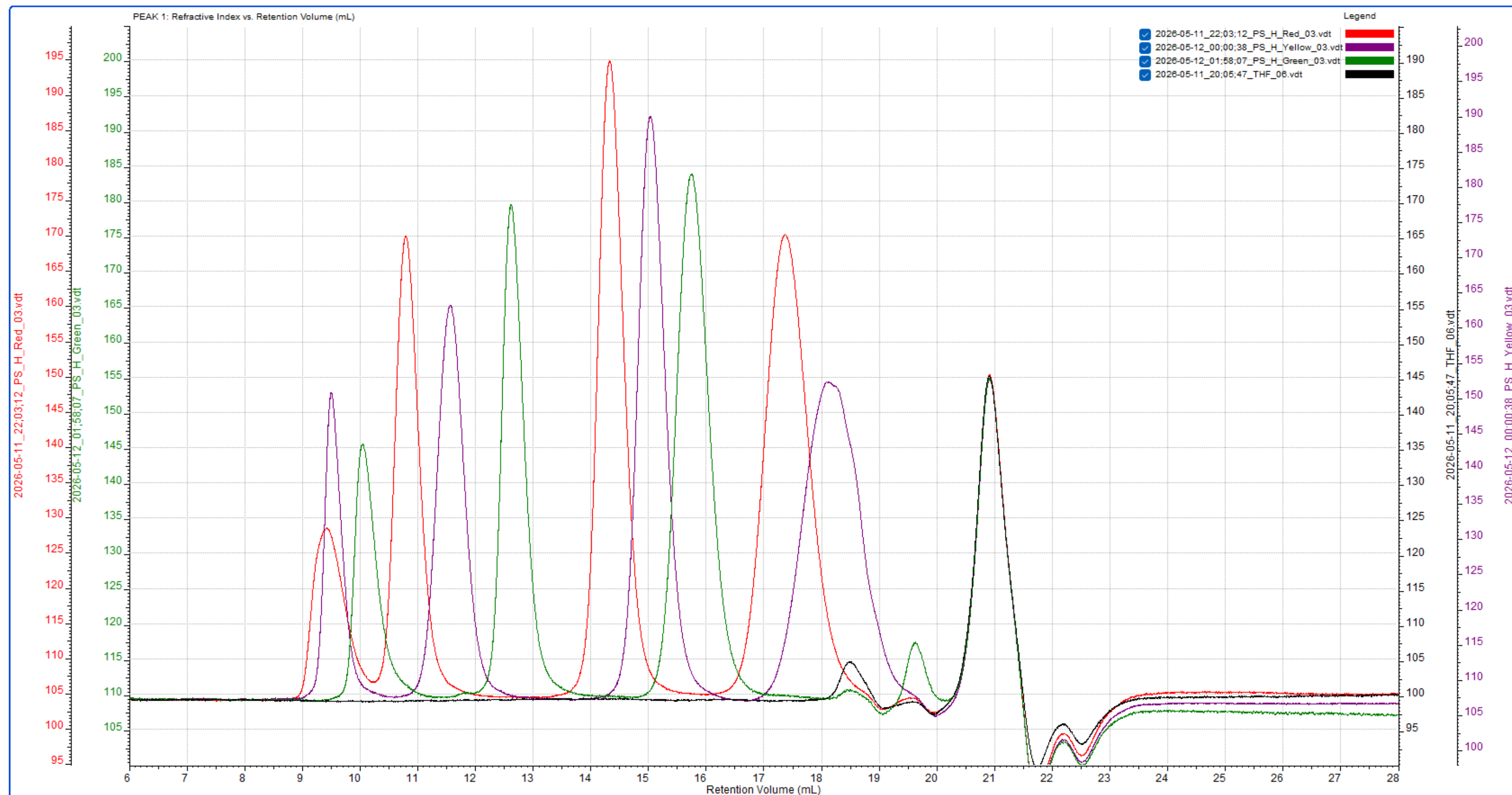
- Molecular weight distributions were characterised for two linear polyester samples, which are used as thermoset resins
- Two samples were submitted
 - **Poly(propylene glycol itaconate) (PPI)**
 - Propylene glycol + itaconic acid
 - **Poly(propylene glycol succinate itaconate) (PPSI)**
 - Propylene glycol + itaconic acid + succinic acid
- Inhibited solution in THF at 50% w/w
 - **Contains hydroquinone (0.06 % w/w) and ethylene glycol (0.12% w/w)**
- The molecular weight distribution was determined by the Gel Permeation Chromatography technique (GPC) in the solvent of THF
 - **Separates species according to hydrodynamic volume in solution (largest first)**
 - **Two different column resolutions were applied to test different size ranges in solution**

Method

- A Viscotek GPCmax/TDA305 system was used
 - **THF as the eluent**
 - **Triple detection system with a concentration detector (refractive index), viscometer and light scattering detector (LALS -7° and RALS -90°)**
 - **Columns and detector were held at a steady temperature of 35°C**
- Two column sets were applied (2 analytical + guard columns in each set)
 - **Low molecular weight resolution (LMWR)**
 - Agilent PLgel 6µm OligoPore
 - **High molecular weight resolution (HMWR)**
 - Agilent PLgel 5µm Mixed-D
- Conventional GPC method was used
 - **Columns calibrated with a series of polystyrene standards**
 - Agilent EasiVial PS-L for low molecular weight resolution
 - Agilent EasiVial PS-H for high molecular weight resolution
 - **Molecular weight values were relative to the elution of polystyrene**
- Samples were diluted to two concentration levels in THF and filtered through an Agilent 0.45 µm (PTFE) membrane
 - **Target 5 and 10 mg/ml**
 - **Actual 2.5 and 5 mg/ml (adjusted for 50% w/w content)**

HMWR – Polystyrene standards

- Elution of EasiVial PS-H standards (3 vials) + THF reference

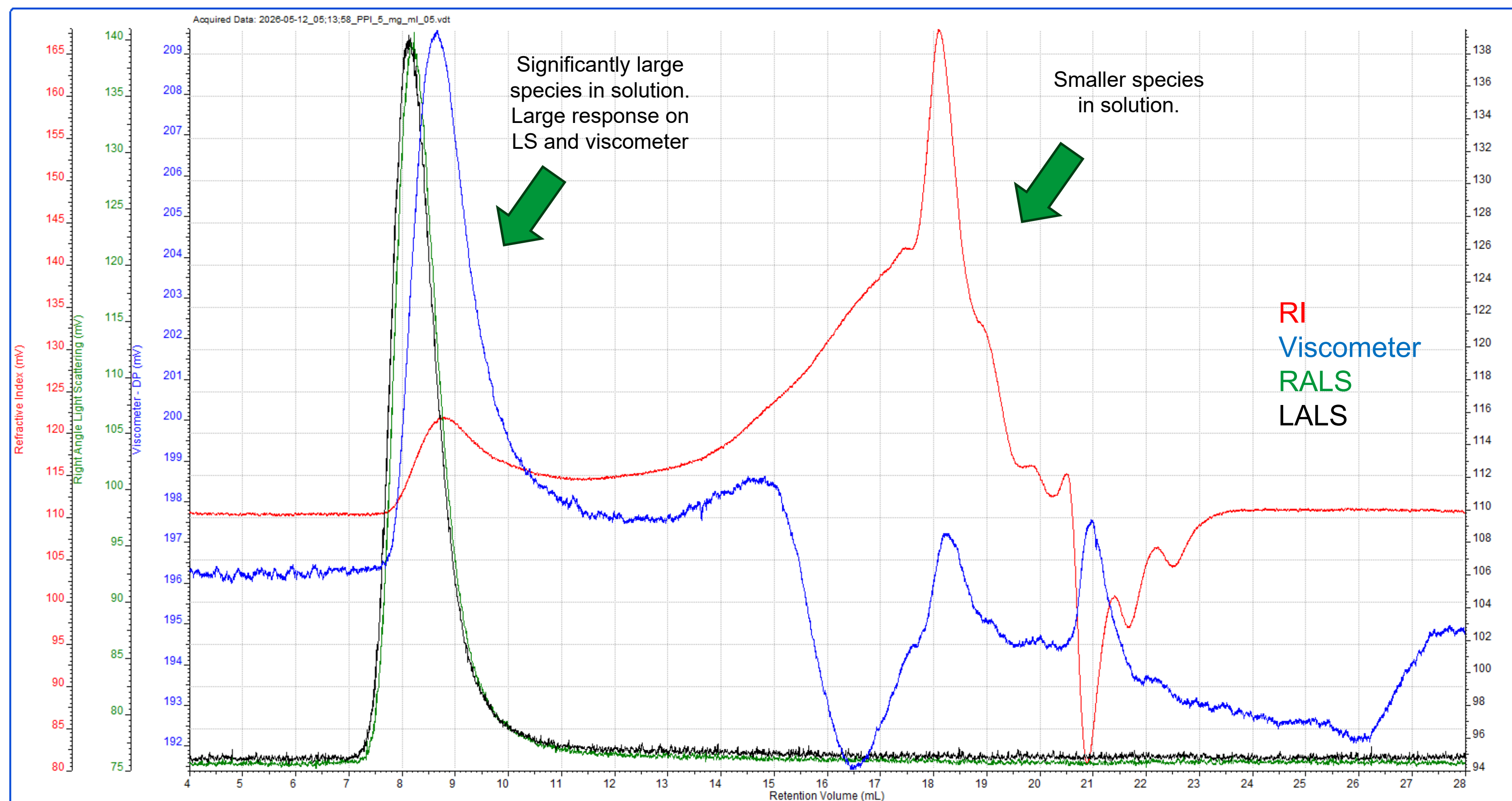


HMWR – Polystyrene standards

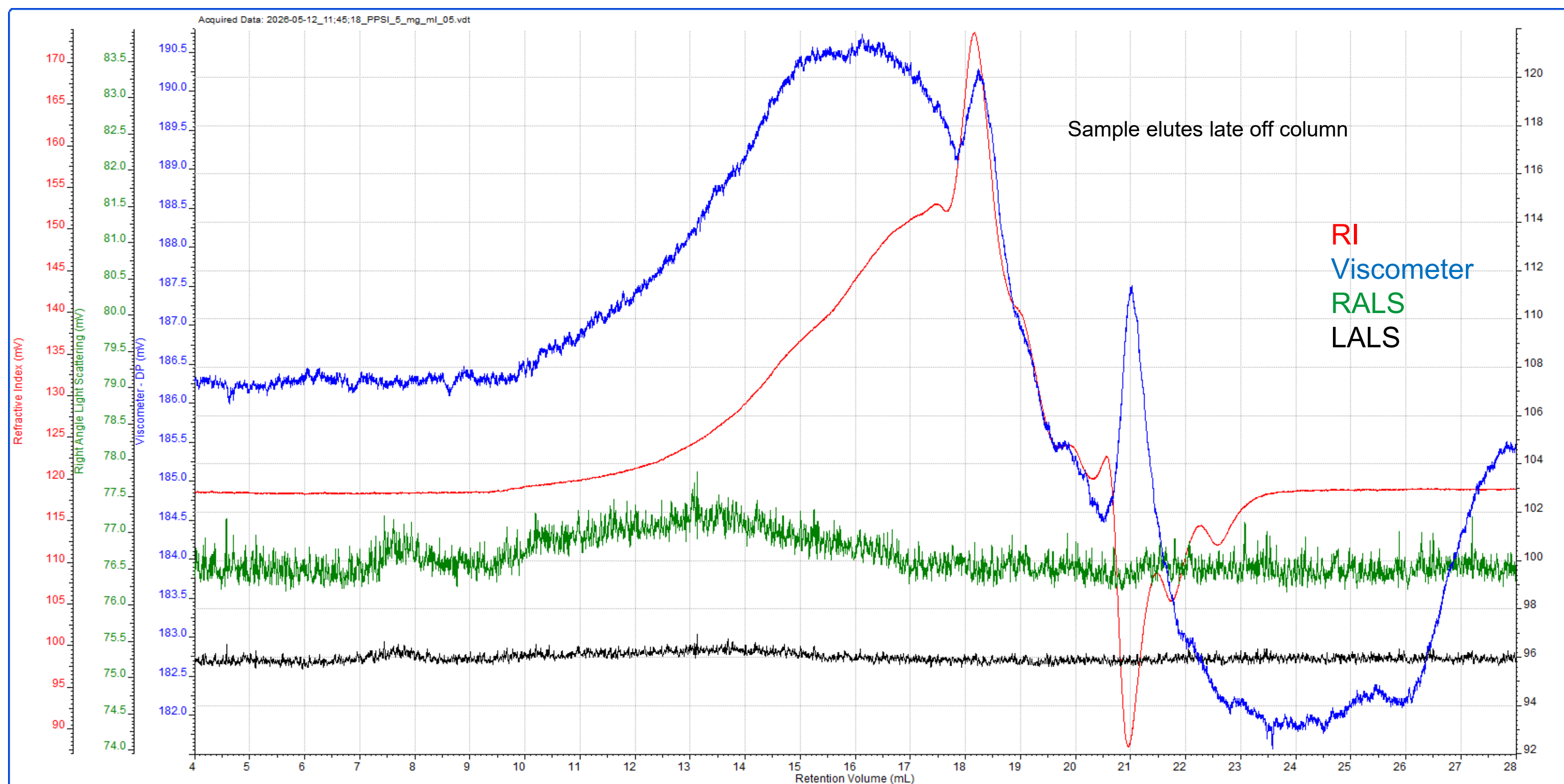
- EasiVial PS-H standards CofA

CERTIFICATE OF ANALYSIS							
Product	Polystyrene High EasiVials (2ml)						
Part Numbers	PL2010-0201, PL2010-0202						
Batch Number	0006680921						
Vial Code	IV (dL/g)	Mw (g/mol) (Light Scattering)	Mn (g/mol)	Mw (g/mol)	Mw/Mn	Mp (g/mol)	Mass/vial (mg)
RED	7.0347	5,725,000	5,855,000	6,275,000	1.07	6,545,000	0.4
	1.1640	420,500	434,000	445,500	1.03	448,500	0.8
	0.1256	18,310	16,720	17,180	1.03	17,120	1.2
	0.0329	1,370	1,100	1,170	1.06	1,180	1.6
YELLOW	4.9513	2,959,000	2,867,000	2,968,000	1.03	3,022,000	0.4
	0.6739	191,900	197,000	204,100	1.04	200,500	0.8
	0.0853	10,060	9,090	9,300	1.02	9,310	1.2
	0.0231	670	555	625	1.13	580	1.6
GREEN	2.2877	1,001,000	1,016,000	1,056,000	1.04	1,069,000	0.4
	0.3424	76,450	72,800	75,000	1.03	75,050	0.8
	0.0600	5,590	4,710	4,860	1.03	4,870	1.2
	-	-	-	-	1.00	162	1.6*

HMWR- PPI - Chromatograms

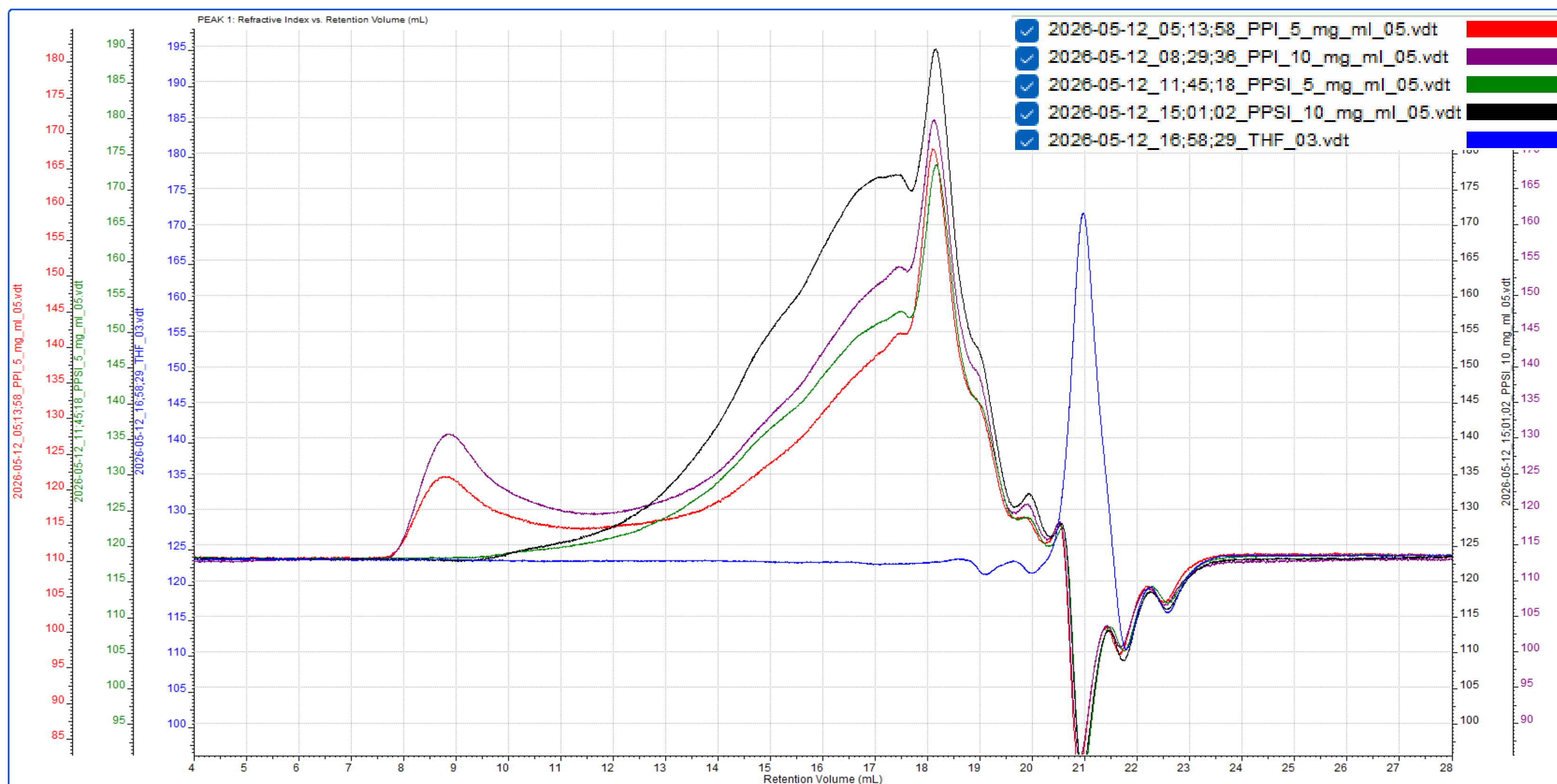


HMWR- PPSI - Chromatograms



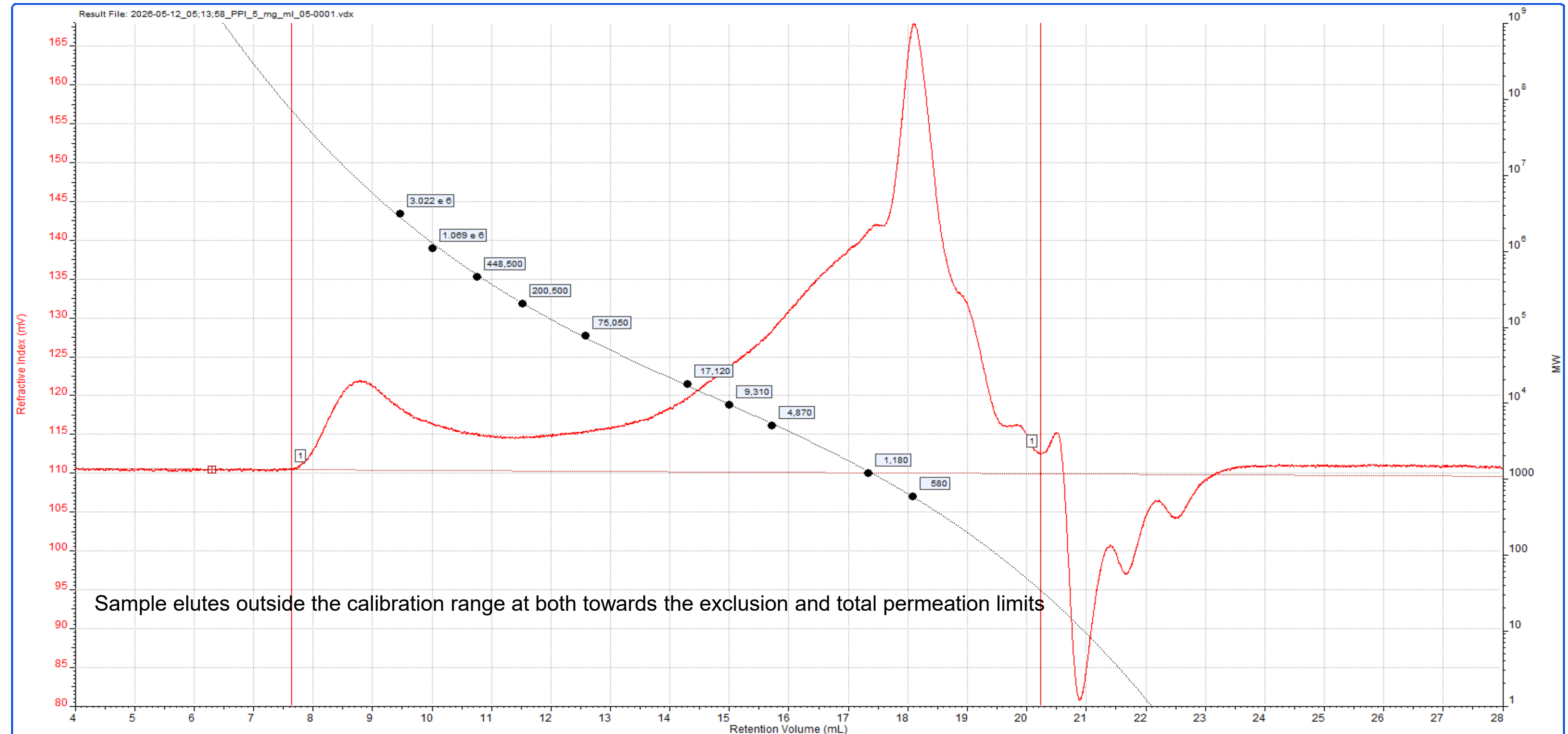
HMWR – Concentration dependence

- Samples injected at 2 different concentration levels



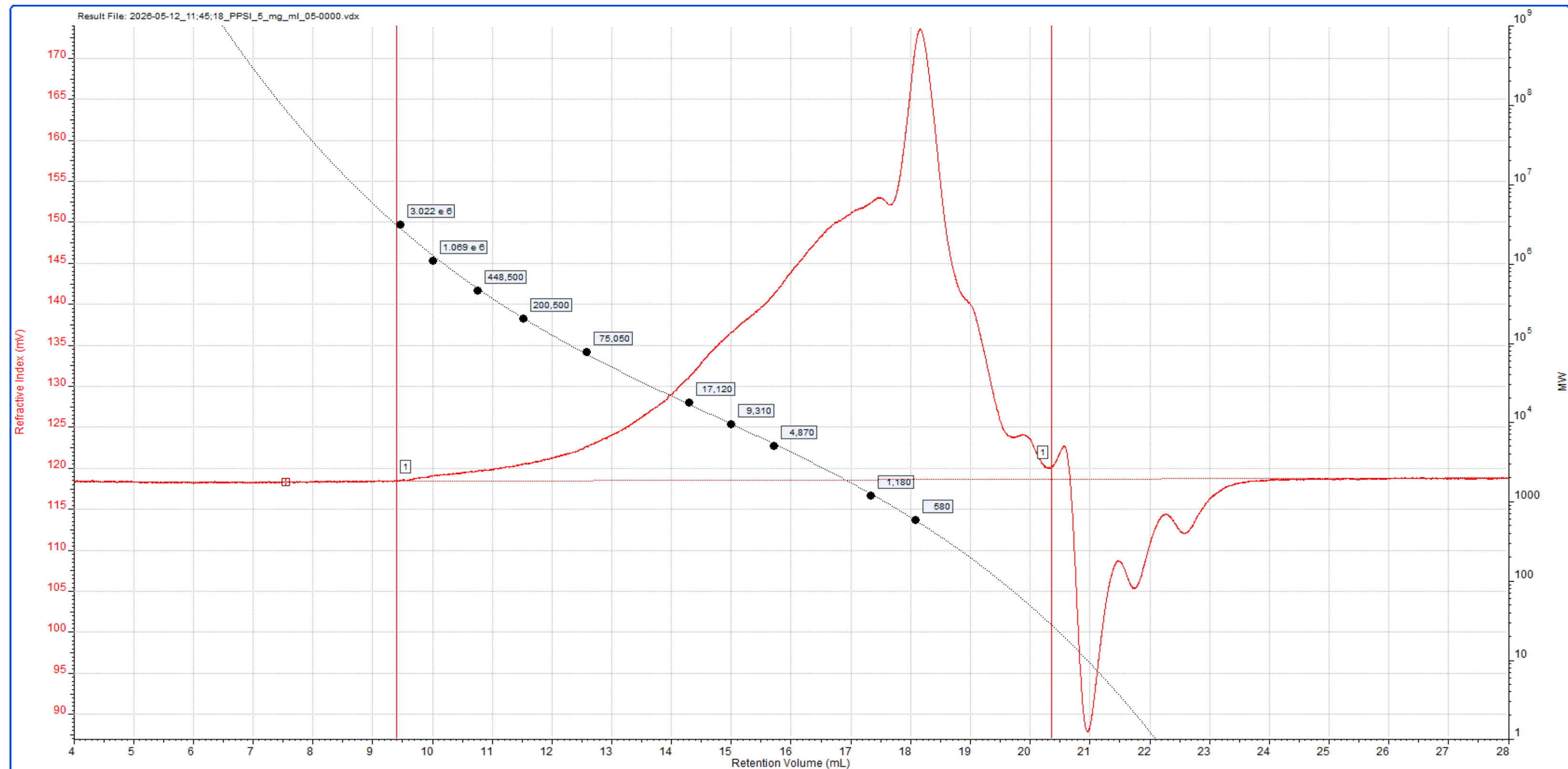
HMWR – Peak markers and baseline

- PPI at 5 mg/ml in THF



HMWR – Peak markers and baseline

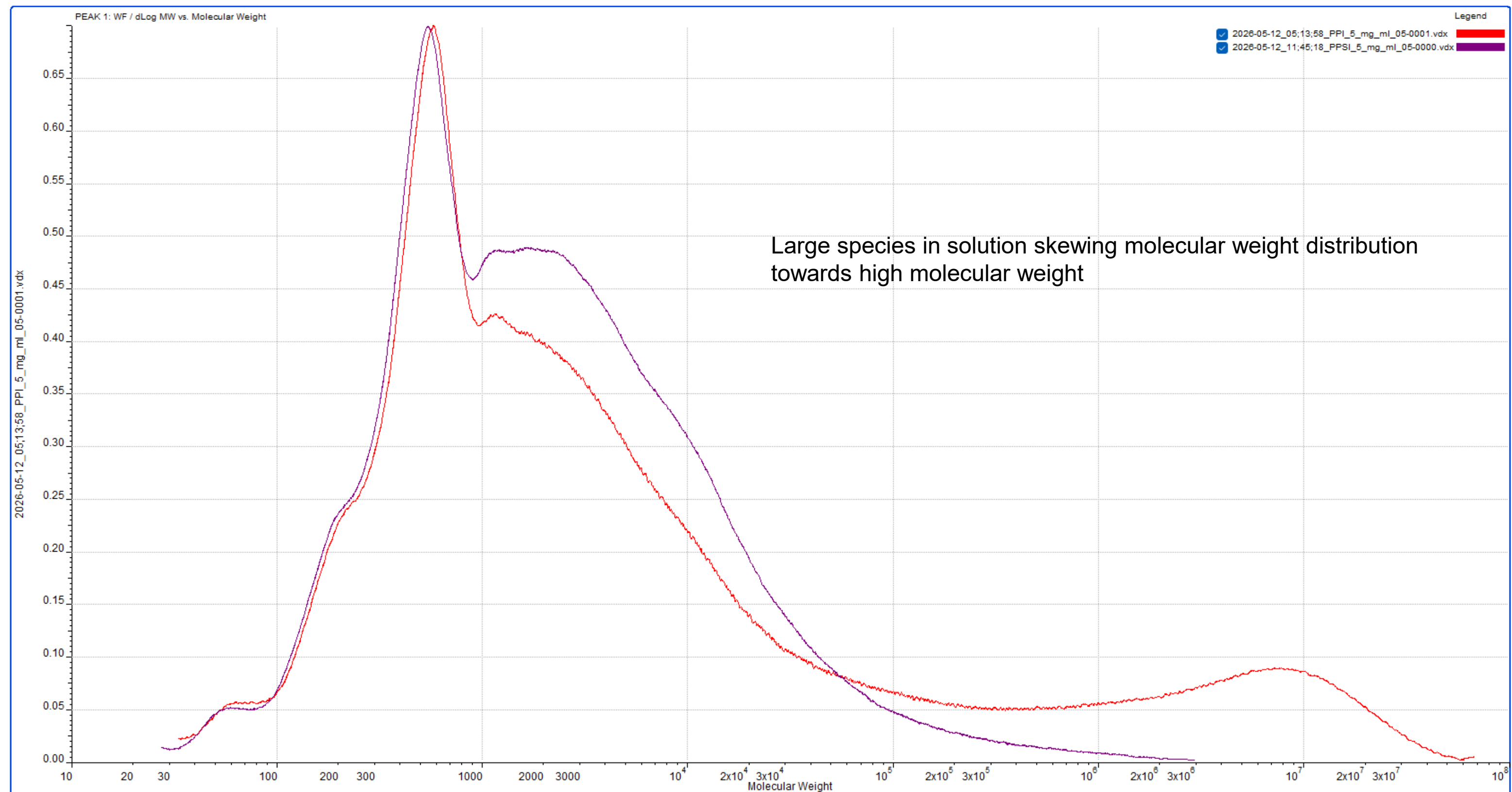
- PPSI at 5 mg/ml in THF



Sample elutes outside the calibration range towards the total permeation limit

HMWR – Molecular weight distribution

- Example overlay of distributions for PPI and PPSI samples



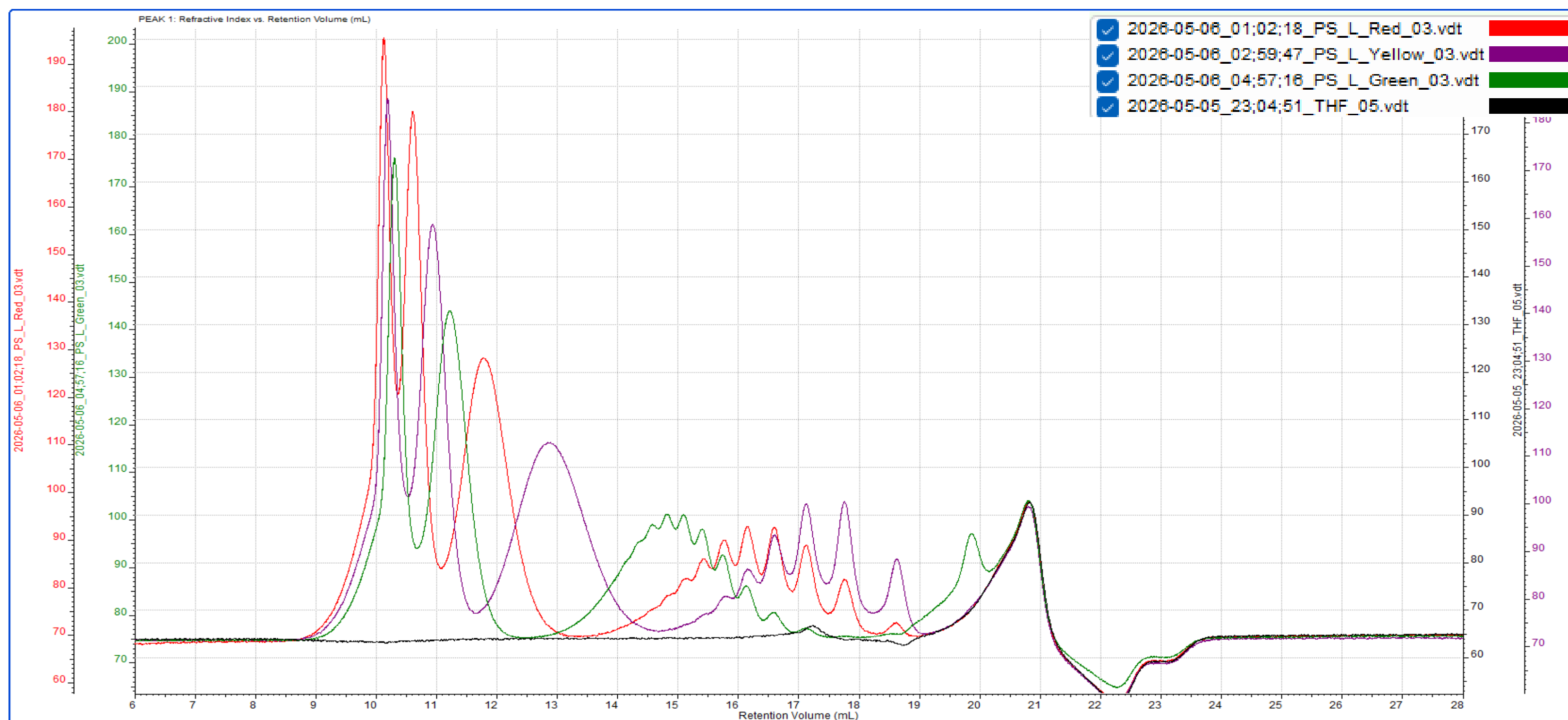
HMWR - Molecular weight data

- Calculated by the conventional GPC method
- Values equivalent to polystyrene in THF
- Each sample/concentration level was injected 5 times (100µL)
- The broad data reflects the range of species as present in solution
 - **Higher values for PPI due to the early eluting peak**
 - **Small concentration dependence was seen (especially with PPI)**
- Higher molecular weight values may be evidence of thermoset reaction starting to proceed

Injection	Sample Id	Acquired	Mw	Mn	Mz
1	PPI 5 mg_ml	05/12/26 02:37:18	973,084	643	1.879 e 7
2	PPI 5 mg_ml	05/12/26 03:16:28	992,613	682	1.757 e 7
3	PPI 5 mg_ml	05/12/26 03:55:38	978,482	688	1.711 e 7
4	PPI 5 mg_ml	05/12/26 04:34:48	965,093	688	1.718 e 7
5	PPI 5 mg_ml	05/12/26 05:13:58	947,092	665	1.715 e 7
6	PPI 10 mg_ml	05/12/26 05:53:06	1.024 e 6	727	1.608 e 7
7	PPI 10 mg_ml	05/12/26 06:32:13	1.031 e 6	741	1.651 e 7
8	PPI 10 mg_ml	05/12/26 07:11:21	1.057 e 6	734	1.757 e 7
9	PPI 10 mg_ml	05/12/26 07:50:29	1.050 e 6	757	1.697 e 7
10	PPI 10 mg_ml	05/12/26 08:29:36	1.015 e 6	743	1.607 e 7
11	PPSI 5 mg_ml	05/12/26 09:08:45	17,598	664	775,179
12	PPSI 5 mg_ml	05/12/26 09:47:53	16,729	651	694,939
13	PPSI 5 mg_ml	05/12/26 10:27:01	17,446	642	699,381
14	PPSI 5 mg_ml	05/12/26 11:06:10	15,378	635	526,099
15	PPSI 5 mg_ml	05/12/26 11:45:18	15,965	624	594,602
16	PPSI 10 mg_ml	05/12/26 12:24:27	17,563	665	543,751
17	PPSI 10 mg_ml	05/12/26 13:03:36	17,220	665	560,694
18	PPSI 10 mg_ml	05/12/26 13:42:44	17,431	658	539,071
19	PPSI 10 mg_ml	05/12/26 14:21:53	17,685	669	569,677
20	PPSI 10 mg_ml	05/12/26 15:01:02	14,506	663	369,739

LMWR – Polystyrene standards

- Oligomeric separation with this column set
- EasiVial PS-L polystyrene standards



LMWR – Polystyrene standards

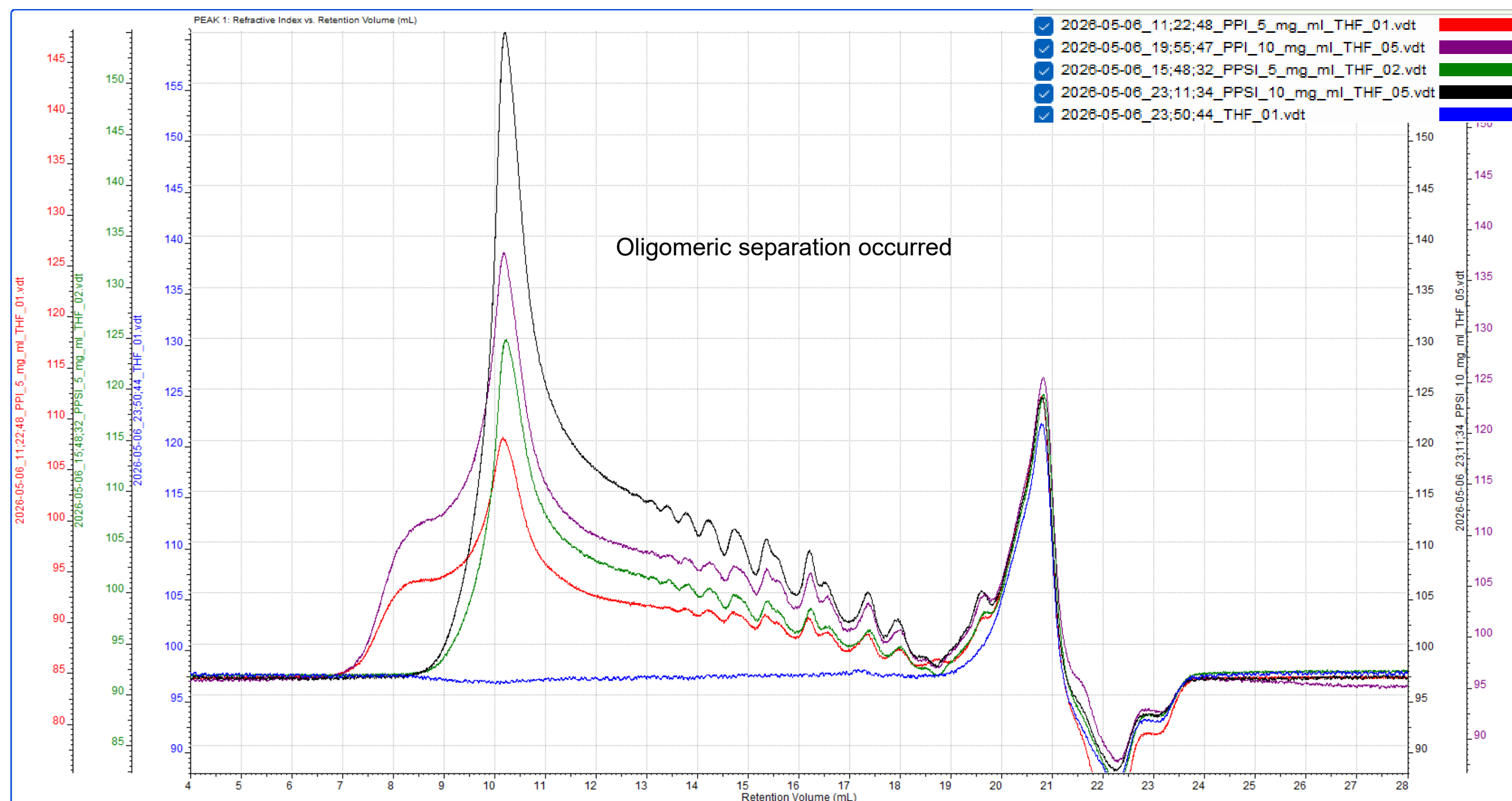
- EasiVial PS-L standards CofA

CERTIFICATE OF ANALYSIS							
Product		Polystyrene Low EasiVials (2ml)					
Part Numbers		PL2010-0401, PL2010-0402					
Batch Number		0006682428					
Vial Code	IV (dL/g)	Mw (g/mol) (Light Scattering)	Mn (g/mol)	Mw (g/mol)	Mw/Mn	Mp (g/mol)	Mass/vial (mg)
RED	0.2543	50,800	46,950	48,900	1.04	49,350	1.0
	0.0853	10,060	9,090	9,300	1.02	9,310	1.3
	0.0503	3,920	3,190	3,310	1.04	3,320	1.5
	0.0231	670	555	625	1.13	580	1.6
YELLOW	0.1757	29,960	27,600	28,250	1.02	28,440	1.0
	0.0691	7,090	6,090	6,260	1.03	6,250	1.3
	0.0393	2,400	1,910	2,000	1.05	1,990	1.5
	0.0262	445	410	450	1.10	370	1.6
GREEN	0.1256	18,310	16,720	17,180	1.03	17,120	1.0
	0.0600	5,590	4,710	4,860	1.03	4,870	1.3
	0.0264	1,100	890	950	1.07	935	1.5
	-	-	-	-	1.00	162	1.6*

* Due to the volatile nature of this constituent weights may vary.

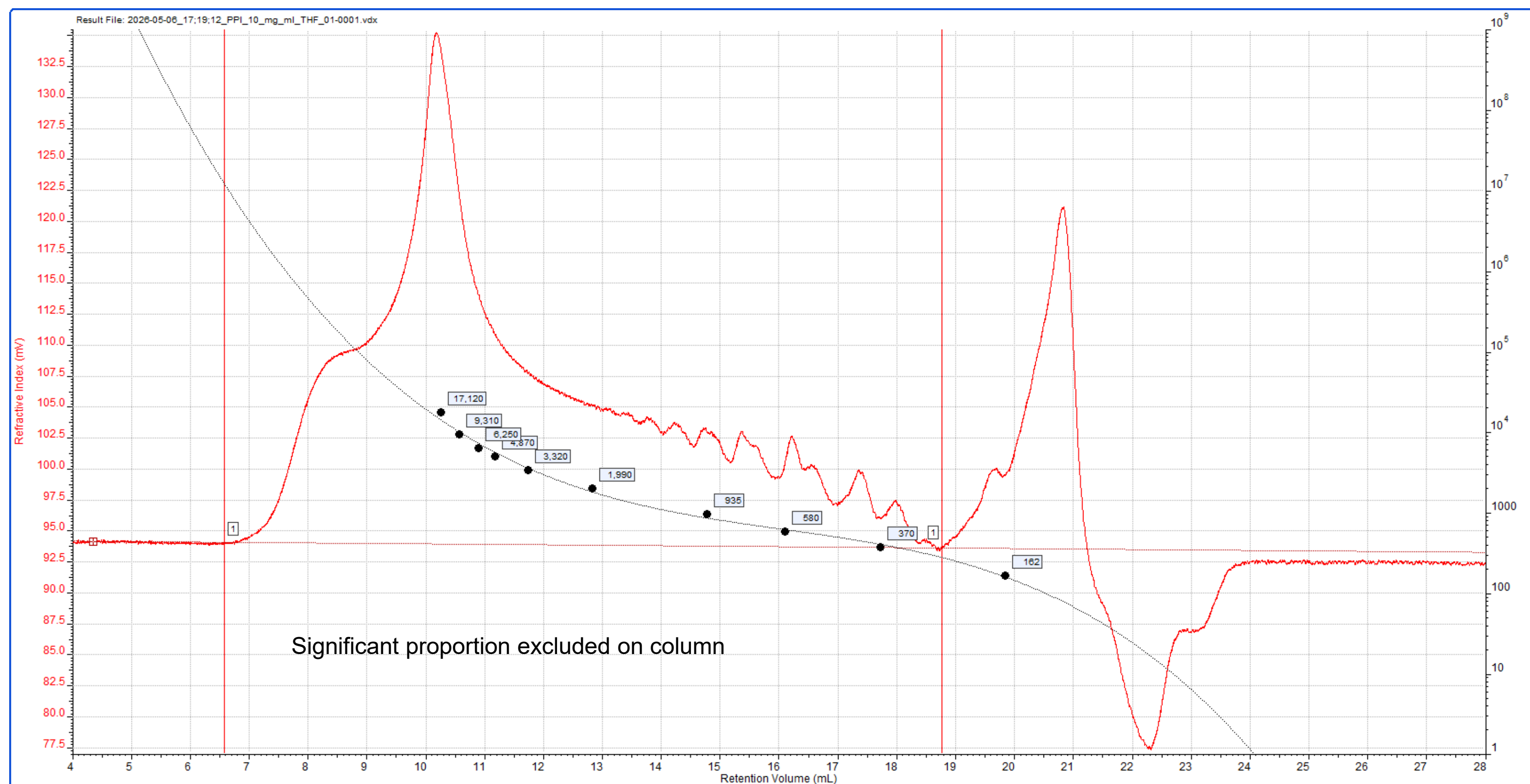
LMWR – Concentration dependence

- Samples injected at 2 different concentration levels



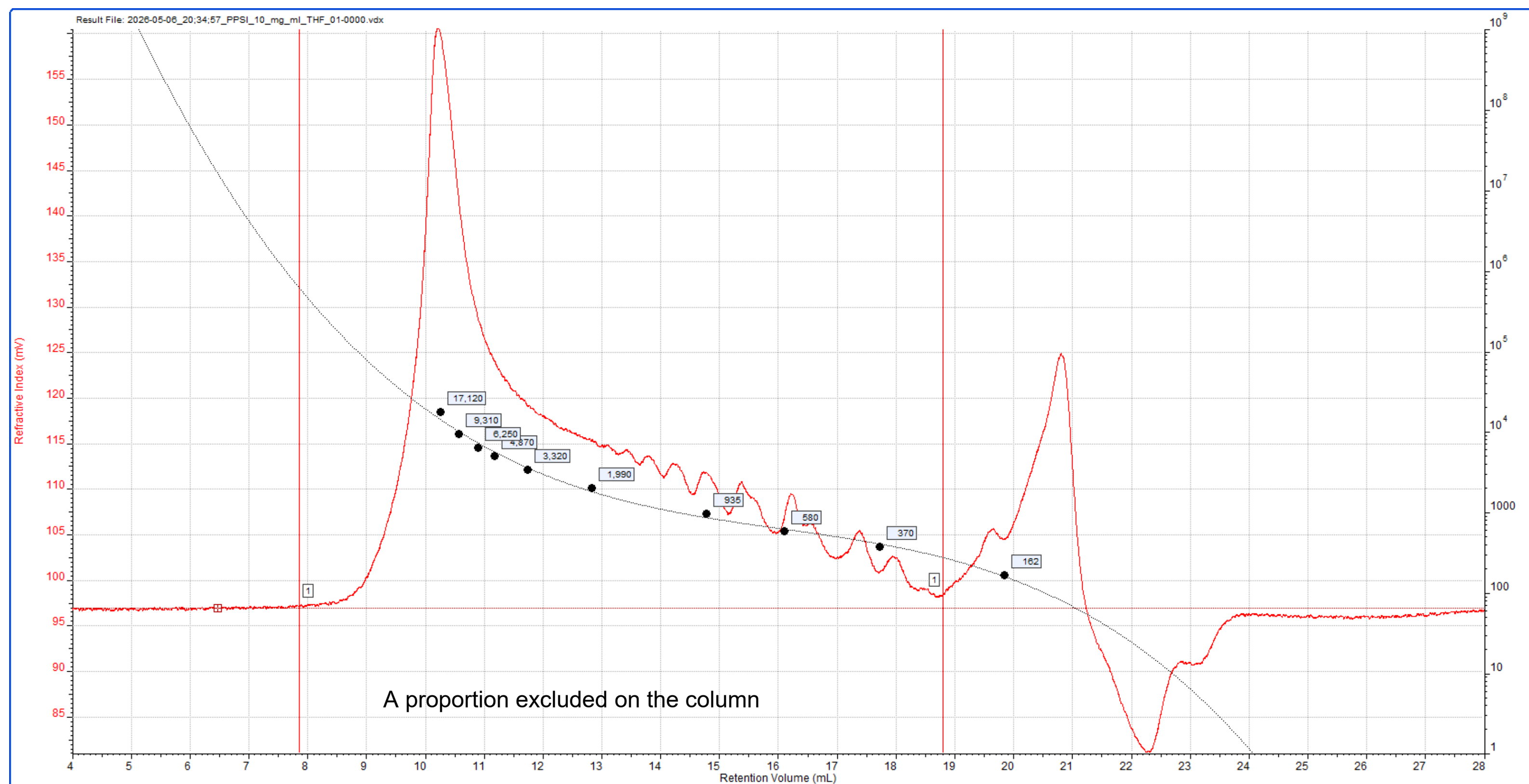
LMWR – PPI chromatogram

- PPI at a concentration of 10 mg/ml in THF



LMWR – PPSI chromatogram

- PPSI at a concentration of 10 mg/ml in THF



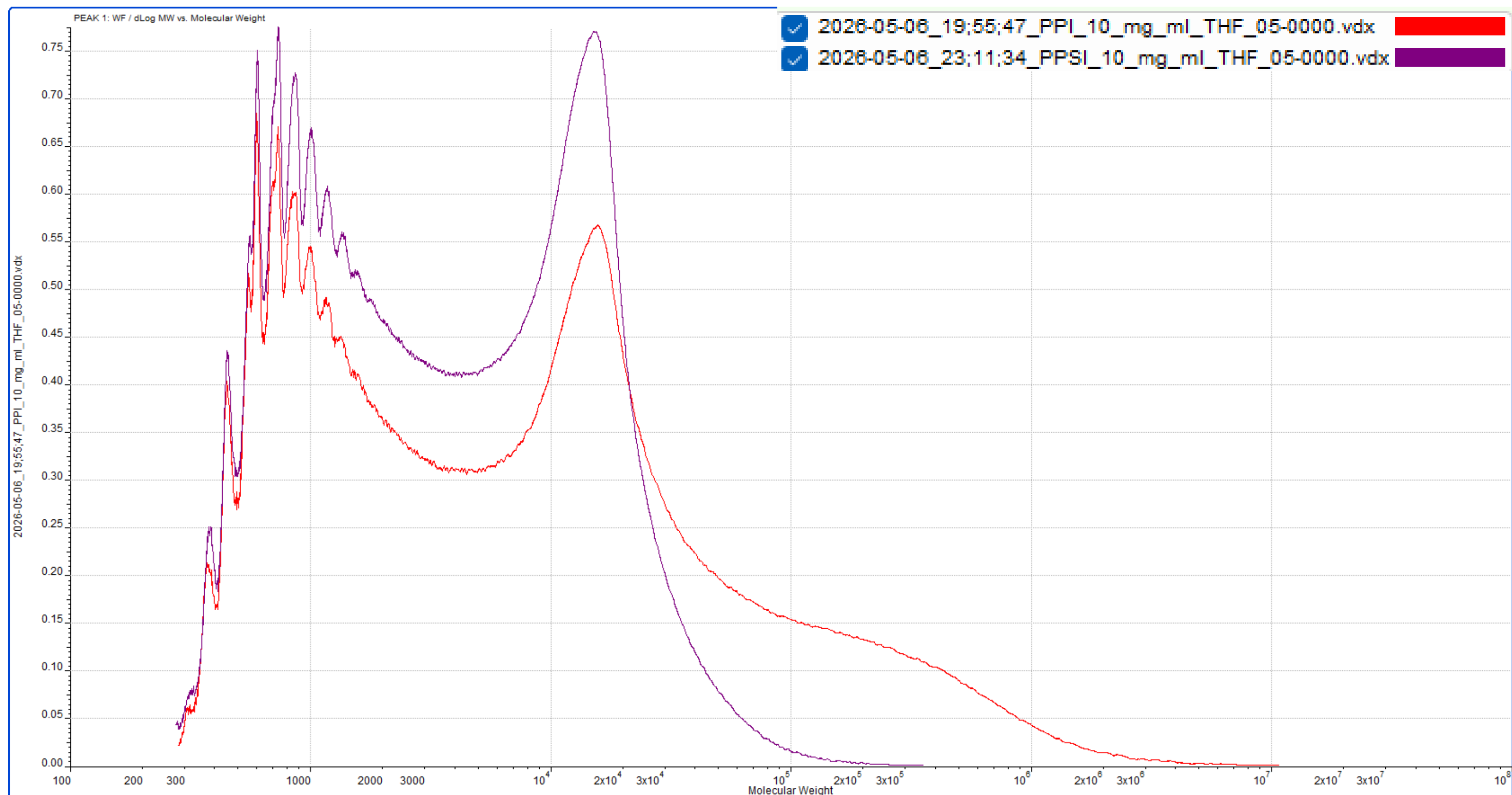
LMWR- Molecular weight data

- Calculated by the conventional GPC method
- Values equivalent to polystyrene in THF
- Each sample/concentration level was injected multiple times (100µL)
- The broad data reflects the range species as present in solution
 - **Higher values for PPI due to the early eluting peak**
 - **Small concentration dependence was seen (especially with PPI)**
- Higher molecular weight values may be evidence of thermoset reaction starting to proceed

Injection	Sample Id	Acquired	Mw	Mn	Mz
1	PPI 5 mg_ml THF	05/06/26 11:22:48	76,037	1,906	1.257 e 6
2	PPI 5 mg_ml THF	05/06/26 13:51:08	71,218	1,969	1.279 e 6
3	PPI 5 mg_ml THF	05/06/26 14:30:16	73,284	1,998	1.377 e 6
4	PPI 10 mg_ml THF	05/06/26 17:19:12	70,017	2,163	1.204 e 6
5	PPI 10 mg_ml THF	05/06/26 17:58:21	62,119	1,943	946,504
6	PPI 10 mg_ml THF	05/06/26 18:37:29	62,969	1,976	1.041 e 6
7	PPI 10 mg_ml THF	05/06/26 19:16:38	69,199	2,293	1.002 e 6
8	PPI 10 mg_ml THF	05/06/26 19:55:47	68,627	1,980	1.319 e 6
9	PPSI 5 mg_ml THF	05/06/26 12:01:56	8,646	1,791	29,358
10	PPSI 5 mg_ml THF	05/06/26 15:09:24	8,275	1,711	27,573
11	PPSI 5 mg_ml THF	05/06/26 15:48:32	8,403	1,772	27,536
12	PPSI 10 mg_ml THF	05/06/26 20:34:57	8,710	1,688	39,655
13	PPSI 10 mg_ml THF	05/06/26 21:14:06	8,467	1,719	30,693
14	PPSI 10 mg_ml THF	05/06/26 21:53:16	8,393	1,714	29,039
15	PPSI 10 mg_ml THF	05/06/26 22:32:25	8,452	1,721	29,272
16	PPSI 10 mg_ml THF	05/06/26 23:11:34	8,450	1,691	30,456

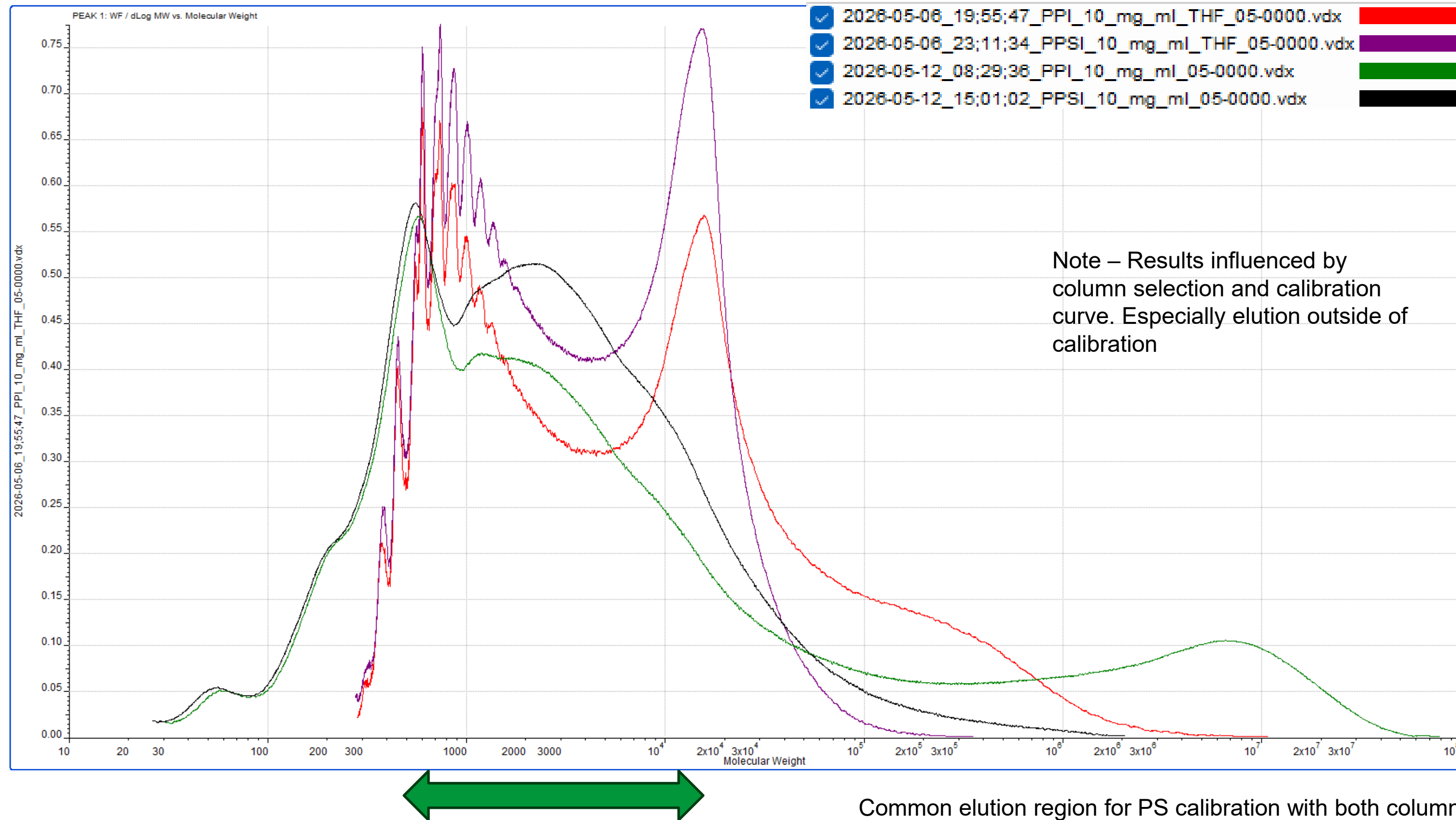
LMWR – Molecular weight distribution

- Example overlay of distributions for PPI and PPSI samples



Combine column resolutions

- Molecular weight distributions for both column resolutions



Summary

- The linear polyester samples were analysed by GPC in the solvent of THF
 - **The samples were higher in molecular weight than expected and were analysed with two column sets**
 - Low and higher molecular weight resolution
 - It was difficult to elute the sample within the calibration range with both column sets due to the broad range of eluting species
- The samples were broad in molecular weight distribution
 - **Possible evidence of the onset of thermoset reaction?**
 - **An early eluting peak was detected for PPI for the presence of significantly larger species in solution**
 - This led to PPI obtaining consistently higher molecular weight values with both column resolutions



Thank You

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